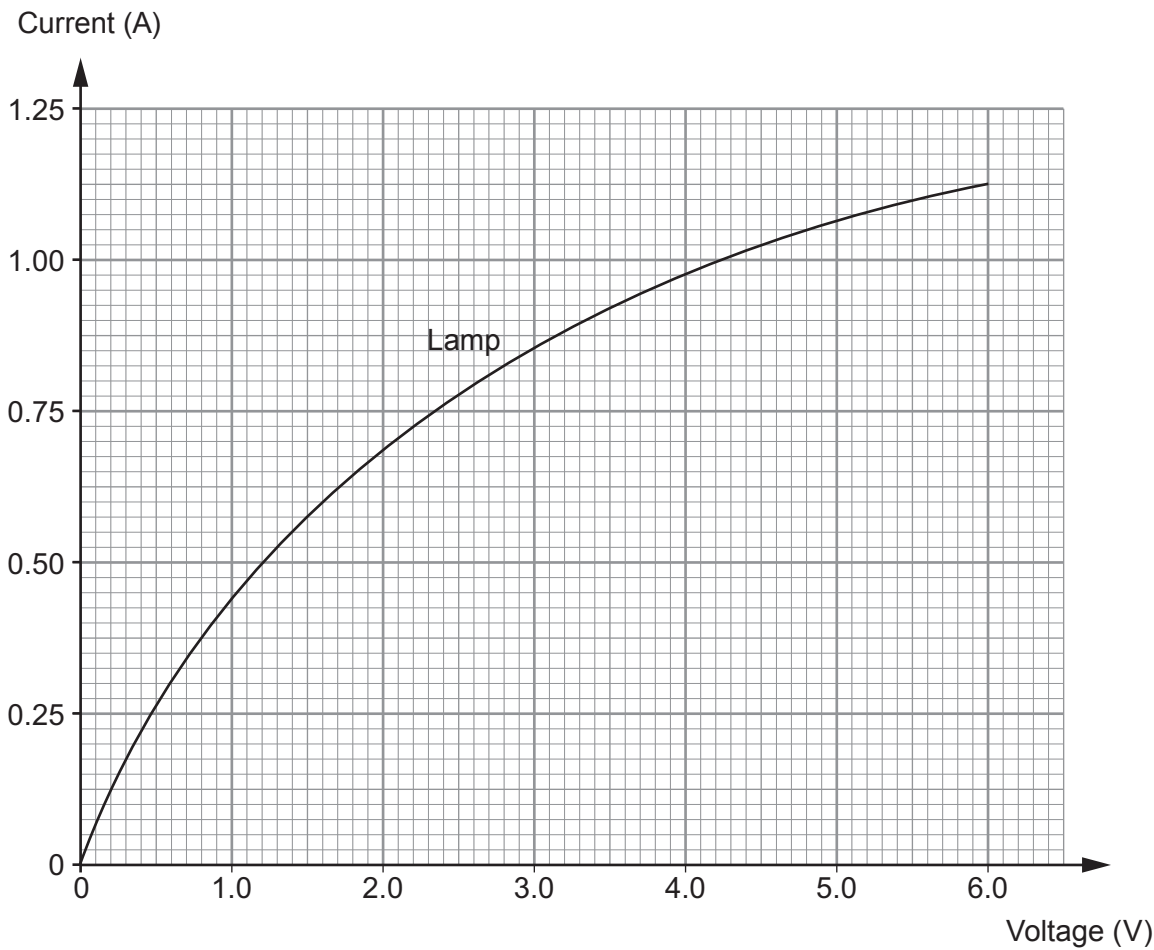


5. In a practical lesson, a group of pupils investigate the voltage across and current through a lamp and a resistor. They draw a graph from their results for the **lamp**. It is shown below.

The current through the resistor is measured and recorded in a table which is shown below.

Voltage across the resistor (V)	1.0	2.0	4.0	6.0
Current through the resistor (A)	0.20	0.40	0.80	1.20

(a) (i) Plot the data for the resistor on the grid below and draw a suitable line. [2]



(ii) Describe how the current through the **lamp** changes as the voltage applied to it is varied. [2]

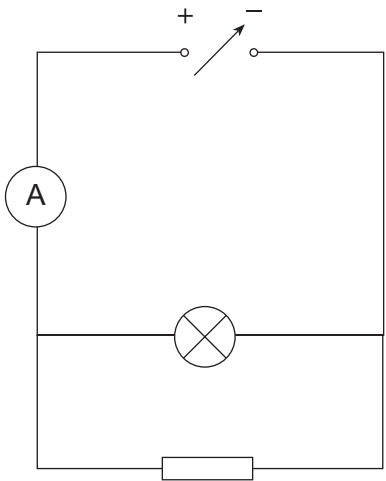
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Examiner
only

- (b) Finally, the group connects the two components in parallel with a variable voltage power supply and an ammeter.



- (i) Use the graph and an equation from page 2 to calculate the resistance of the lamp at 3.0 V. [2]

Resistance = Ω

- (ii) Calculate the ammeter reading at 3.0 V. [2]

Current = A

- (iii) Calculate the total resistance of the circuit at 3.0 V. [2]

Resistance = Ω

Examiner
only

(iv) Explain, in terms of current, why the whole circuit has less resistance than the resistance of either component. [2]

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(v) The voltage supply is increased from 0V to 6.0V. Explain how **the graph shows** that the lamp has a bigger resistance than the resistor at 6.0V. [2]

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(vi) Use the graph and without calculations but giving clear reasoning, compare the powers of the lamp and the resistor between 0 and 6.0V. [2]

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